Sporothrix Genome DataBase Update and Whole Phylogenomic Analysis

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INTRODUCTION

The Sporothrix Genome DataBase (SGDB) was established in 2020 as the first web-based platform dedicated to collecting DNA/RNA-seq data and gene models of the fungal pathogen *Sporothrix schenckii* (1). With the growing number of *Sporothrix* genomes sequenced, more genomic data became available and suitable for their integration into the SGDB.

The current version of the database, available at http://sporothrixgenomedatabase.unime.it/, went through several updates since 2020, substantially expanding its content to include 7 different complete *Sporothrix* genomes and their gene annotations with the possibility of downloading and visualizing them via the JavaScript-based genome browser (JBrowse v.1.12.3). The new genomes were retrieved from the National Center for Biotechnology Information (NCBI) and include: (i) *S. curviconia* CBS 959.73; (ii) *S. globosa* CBS 120340; (iii) *S. globosa* SS01; (iv) *S. phasma* CBS 119588; (v) *S. protearum* CBS 116654; and (vi) *S. pallida* SPA8. Based on genomic data available in both NCBI and SGDB databases, we decided to investigate the evolutionary relationships of 31 *Sporothrix* strains sequenced so far and compare results with the current known taxonomy of the genus. Phylogenomic analysis was performed using both mitochondrial and whole genome assemblies discarding from the first inference those strains whose mitochondrial sequences were found to be misassembled into nuclear contigs/scaffolds.

SGDB
Sporothrix Genome DataBase

Sporothrix Genome DataBase

We are pleased to announce the launch of the Sporothrix Genome DataBase, an online resource that provides gene, protein and genomic sequence data for the dimorphic fungus Sporothrix schenckii and related species. Taking advantage of next-generation sequencing technology, the Sporothrix Genome DataBase offers an integrated tool set for accessing, analyzing and exploring genomics data.

MATERIALS AND METHODS

The database update involved 6 species:
S. pallida (GCA_000710705.2) (5),
S. globosa (GCA_001630435.1 and *GCA_001630445.1*) (4),
S. phasma (*GCA_016097075.2*),
S. curviconia (*GCA_016097085.2*),
S. variecibatus (*GCA_016097105.2*),
S. protearum (*GCA_016097115.2*) (3).

The genomes and the corresponding annotations (*.gff3) were uploaded to the database (SGDB). Phylogenetic analysis using Jolytree (v2.1.211019ac) (8) was performed on all *Sporothrix* strains deposited in Genbank (6). In order to perform the phylogenetic analysis at mitochondrial level, we first identified single contig/scaffold sequences corresponding to the complete mitogenome with Exonerate software (version 2.4.0) (7): the whole-genomes, downloaded from Genbank were used as the targets, while the *S. phasma* (CBS 119588) mitogenome was used as query. Each mitogenome not misassembled into nuclear contigs/scaffolds was included in the analysis. The phylogenomic analysis using the whole nuclear genomes, involved a total of 31 *Sporothrix* strains (Fig. B). On the other hand, phylogeny based on mitochondrial sequence data (Fig. C), involved only 21 *Sporothrix* assemblies, as strains with incorrectly assembled mitogenomes were excluded from the analysis.

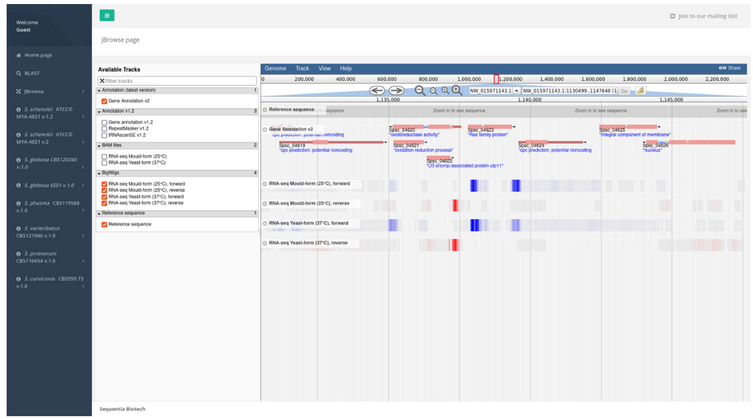
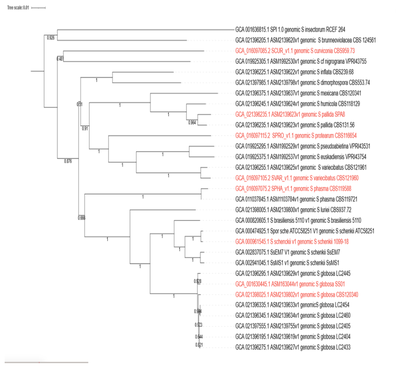


Fig. A : Screenshot of the JBrowse integrated into the SGDB. All *Sporothrix* strains, currently deposited in the database, are shown on the left.



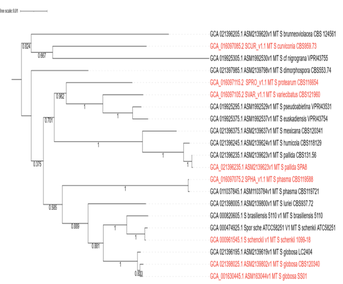


Fig. B: Phylogenetic tree of all 31 *Sporothrix* whole nuclear genomes.
Fig. C: Phylogenetic tree of all complete *Sporothrix* mitogenomes.

RESULTS AND CONCLUSIONS

The SGDB is now updated with new mitogenomes and gene annotations, storing data for a total 7 *Sporothrix* species (Fig. A). The BLAST section now allows the search against specie-specific as well as whole genomes, transcripts and proteins sequence data. The taxonomic relationships among the 32 *Sporothrix* strains included in this study showed solid bootstrap values, suggesting that both nuclear and mitochondrial inferences suitable in describing the evolutionary history of these fungi. Comparing our result with literature data, we did not find any substantial difference, supporting the use of both mitogenome and/or nuclear genome as valuable markers for species discrimination and identification within the *Sporothrix* genus.

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